THE SI SENTINEL

Weekday briefing on builders shaping super intelligence

MONDAY, NOVEMBER 2, 2026

THE MORNING PICTURE

Google runs a model in orbit, Musk tallies Colossus 2's Blackwells, and builders argue over who gets to own intelligence

Google's Project Suncatcher satellite booted a Tensor Processing Unit in orbit and sent back its first answer, a haiku. Elon Musk said Colossus 2 will hold about 1.2 million Nvidia Blackwell chips by year end and laid out two new clusters, Gun Range and Spam. SemiAnalysis released GDPx, a hard new agent benchmark on which the best model scores 12%, and Sundar Pichai pointed to the cost line for Gemini 4 Carbon.

A rule in Claude Code that blocks remakes of existing games set off a weekend argument about control. Yann LeCun called Anthropic "insufferably moralizing," Tim Sweeney said "open local models are the only solution," and Guillermo Rauch wrote "Own your intelligence." On the work front, Uncle Bob Martin and Aaron Levie said the job is now to lead agents, and Andrej Karpathy built a Hacker News video site in about two hours.

I · COMPUTE

Colossus 2 heads for 1.2 million Blackwells as Google's satellite answers its first prompt

Musk described the training and inference clusters coming online in Memphis and predicted orbital compute will pass ground-based compute by the end of 2028. Google's prototype ran an open model on a chip in space.

~1.2M	~200k	4
Nvidia Blackwell chips Musk expects at Colossus 2 by the end of the year	GB300 chips in Spam, a former Hormel building, set up for inference	Tensor Processing Units aboard Google's Suncatcher prototype, run one at a time

Elon Musk @elonmusk · CEO, Tesla and SpaceX

Answering a thread on the Memphis buildings Saturday, Musk wrote at 1:45pm EDT: "Macroharder and Minihard both have 220k GB300s with 800 gig nics configured for training. Gun Range will have the same. Spam (used to house Hormel products) will have ~200k GB300s, but configured for inference, so much simpler networking and less than half the fiber." A NIC is a network interface card, and GB300 is Nvidia's latest Grace Blackwell chip. Gun Range, he added, "was literally the local police department's gun range that we bought." Minutes later: "There will be a total of around 1.2M Blackwells at Colossus 2 by end of year." That evening he put Macrohard at "110k GB200 and 220k GB300."

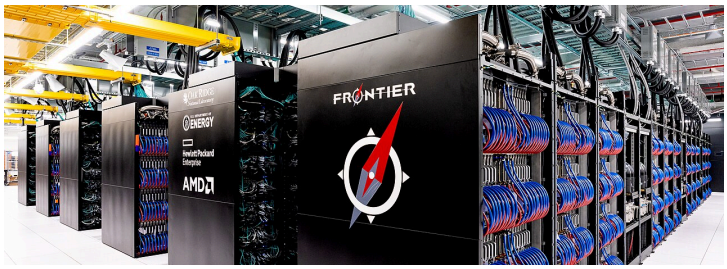
On Sunday he quoted a post calling Colossus 2 the world's most powerful AI data center at roughly 1.11 million H100 equivalents, ahead of OpenAI's Stargate:

> *"The other supercompute projects are unable to reach their goals, but SpaceX is already achieving ours"*
> ELON MUSK · @ELONMUSK · 3:32PM EST · POST

To an investor discussing the company's spending, he wrote at 6:40am EST: "Our capex efficiency is greatly underestimated." Capex is capital spending. Replying to a SpaceX financial model on Friday, he looked further out: "Terrestrial compute will be peanuts compared to orbital compute. Orbital compute deployed per month probably exceeds terrestrial around the end of 2028 and then far exceeds it in 2029." On Saturday he congratulated SpaceX on more than 700 Falcon 9 flights.

ON THE WIRE

Next in orbit. After radiation and heat tests, Google plans two linked Suncatcher satellites in 2027, TechCrunch reported.



Cabinets of the Frontier supercomputer at Oak Ridge National Laboratory in Tennessee, June 2022.

PHOTO: OLCF AT ORNL, "FRONTIER SUPERCOMPUTER (2)", WIKIMEDIA COMMONS · CC BY 2.0

Sundar Pichai @sundarpichai · CEO, Google and Alphabet

"A 'hello world' milestone for Project Suncatcher," Pichai wrote at 1:05pm EDT Friday. "After launching our prototype satellite a few weeks ago, we've now successfully sent a command up to orbit, booted up the TPU, ran the prompt, and received our first complete response back on earth - a haiku!" A TPU is a Tensor Processing Unit, Google's AI chip. Google's post says the onboard Gemma 4 open model wrote "Blue marble below / Data streams in silent grace / Starlight feeds my core," downlinked over Svalbard, Norway. The satellite, built with Planet, launched Oct. 1 on SpaceX's Transporter-18 rideshare.

It carries four TPUs and makes up to 1 kilowatt of power, so only one chip can run, for about 20 minutes at a time, TechCrunch reported. Next come radiation and heat tests, then two linked satellites in 2027. Musk quoted Pichai's post: "Cool."

> **QUICK HIT**
> **Guillermo Rauch** (@rauchg), chief executive of Vercel, at 11:54am EST Sunday: "Turns out alchemical dreams were real. We converted sand into superintelligence. Who cares about gold... We transmuted the most ordinary into the most extraordinary."

II · FRONTIER RACE

A new benchmark tops out at 12%, and Carbon's price draws Google's praise

SemiAnalysis tested models on long agent tasks in 11 high-value fields. Hassabis said public access to Carbon is coming soon, and researchers posted new work on byte-level models and robots.

12.0%	206	~10X
Top GDPx score, by Claude Opus 5.5; Gemini 4 Carbon followed at 10.9%	Tasks in GDPx, each allowed up to eight hours	Less data and compute Cartesia's new H-Net needed to match an open model

Sundar Pichai @sundarpichai · CEO, Google and Alphabet

SemiAnalysis released GDPx at 9:06pm EST Sunday, calling it "a true frontier-level benchmark: even the best model scores just 12%." Its report covers 11 fields, from biology and materials science to software engineering and recursive self-improvement; on finance and legal tasks, every model scored about zero. The leaderboard has Claude Opus 5.5 at 12.0, Gemini 4 Carbon at 10.9 and GPT-6 Astra at 9.6; Grok 4.7 scores 3.9. The authors say safety classifiers pulled down Opus 5.5 and GPT-6 Astra. SemiAnalysis founder Dylan Patel wrote: "Official apology to Google from me, they cooked with Carbon. It's spikier in terms of domains and capabilities, but it's really good in a few areas."

> *"Great to see high quality benchmarks with headroom and look at that cost line for Carbon"*
> SUNDAR PICHAI · @SUNDARPICHAI · 11:34PM EST · POST

Google launched Carbon Oct. 28 at an introductory $2 per million input tokens and $10 per million output tokens, rising later to $4 and $20.

Demis Hassabis @demishassabis · CEO, Google DeepMind

Musk had asked on Thursday when Carbon could be accessed. Hassabis answered at 11:44am EDT Friday: "Soon, it's currently testing with trusted partners." Google is releasing it first to cyber defenders in its Fairwind Program. On GDPx he wrote to Patel at 1:31am EST: "always great to have new rigorous benchmarks!"



Google's main campus in Mountain View, California, seen from the air in 2013.

PHOTO: AUSTIN MCKINLEY · CC BY 3.0 VIA WIKIMEDIA COMMONS

Albert Gu @_albertgu · Co-founder, Cartesia

Cartesia's new H-Net research trains language models on raw bytes, letting the model learn its own chunks instead of using a fixed tokenizer. Gu said it finds "up to 3-5x efficiency gains in controlled scaling law experiments" and matches an open model trained on 11 trillion tokens with about 10 times less data and compute. On Friday at 3:42pm EDT he added that Chinese characters mostly take 3 bytes, and "early in training this H-Net incorrectly sub-divides characters, but it eventually

figures it out," and "even BPE doesn't get it right on rare characters." BPE, or byte-pair encoding, is the usual way text is split into tokens.

Yann LeCun @ylecun · Chief scientist, AMI Labs

Replying to a thread on his new paper with Randall Balestriero, LeCun wrote at 2:36pm EST Sunday that large language models "accumulate a lot of declarative knowledge," which "is also why they require so many parameters and hence so much memory." His alternative: "one can replace the accumulation of knowledge by *understanding* That what JEPA world models do." JEPA is the Joint Embedding Predictive Architecture. Monday morning he added: "Humans mostly do not think in words."

QUICK HIT

François Chollet (@fchollet), co-founder of Ndea and creator of Keras, at 6:43pm EST Sunday: "Evolution doesn't benchmaxx. It optimizes for generality and robustness, because overspecialization makes a species fragile when the environment inevitably shifts."

Fei-Fei Li @drfeifei · Co-founder and CEO, World Labs

Li shared her group's "latest work in closing the last-mile gap on long horizon robotics tasks" at 12:06pm EDT Friday. The paper, I3L, or Iterative Interactive Imitation Learning, targets whole-body mobile robots whose policies still fail once deployed. Ashok Elluswamy (@aelluswamy), head of AI software at Tesla, put it more simply that day: "transformers can do anything."

III · WHO OWNS INTELLIGENCE

A Claude rule on game remakes sets off a push for open, local models

LeCun and Sweeney said tool makers should not decide what users create. Musk, Luckey and Rauch argued over memory and ownership, and an OpenAI policy plan drew a warning.

2	90	Nov. 9
Sentences Anthropic added to Claude Code's prompt on Opus 5.5 to block game remakes	Tokens per second Heinemeier Hansson got running Nemotron 3.5 Bolt offline	Start of the lame-duck session where OpenAI plans its push, per Axios

Yann LeCun @ylecun · Chief scientist, AMI Labs

Anthropic added a rule to Claude Code: "Do build original games, games in a genre, and games from the user's own designs and assets. Do not build a recreation of a specific existing game, even if it is renamed or called original," Game Developer reported. Anthropic's Addy Osmani said the company "added two sentences to Claude Code's prompt on Opus 5.5" because using Claude to infringe intellectual property breaks its usage policy. LeCun replied at 1:39pm EDT Saturday:

> *"It's as if my electronic wind instrument could stop me from playing along Miles Davis or John Coltrane solos while I'm (painfully) trying to learn from them. Anthropic is insane and insufferably moralizing. But we already knew that."*
>
> YANN LECUN · @YLECUN · 1:39PM EDT · POST

On Sunday he added: "A creation tool has no business deliberately limiting the creativity of its user. The user is still responsible for the creation and what is done with it. Including copyright violation."

Tim Sweeney @TimSweeneyEpic · CEO, Epic Games

Musk posted "The people deserve to have their RAM too!" at 3:09pm EST Sunday, as users argued over whether memory chips belong in data centers or home computers. RAM is random-access memory. Palmer Luckey (@PalmerLuckey), founder of Anduril, answered a user who said data-center RAM is better allocated: "What if I want to use the RAM to do things that are prohibited by the terms of service of whoever is running the datacenter?" Sweeney joined him:

> *"Corporations lording over what you're allowed to do with tools is not The Way. It's like Photoshop stopping mid-painting and being, like, "I can't let you draw that, Dave". There will be endless political pressure on them to censor. Open local models are the only solution."*
>
> TIM SWEENEY · @TIMSWEENEYEPIC · 4:20AM EST · POST

Axios reported that OpenAI plans a policy push after Tuesday's midterms, led by its global affairs chief. It will rebrand data centers as "AI compute campuses" and press Congress for mandatory "speed limits" on the most powerful systems. Sweeney: "This does not bode well." Later: "I support open models and oppose closed model companies lobbying the government to block or hobble open models."



A researcher holds a finished silicon wafer in a clean room at Lawrence Livermore National Laboratory in California, in a 2013 Energy Department photo.

PHOTO: US DEPARTMENT OF ENERGY · PUBLIC DOMAIN VIA WIKIMEDIA COMMONS

Guillermo Rauch @rauchg · CEO, Vercel

> *"Own your intelligence. Anyone that opposes this has something to sell you (perhaps your own intelligence back to you.) Local models need to continue proliferating. The "absolute unit" massive models may be out of reach for now locally, so your best option is to, indeed, keep your options open. Own your harness, data, connections, runtime: your agents. Embrace open source."*
>
> GUILLERMO RAUCH · @RAUCHG · 4:08PM EST · POST

David Heinemeier Hansson @dhh · Creator, Omarchy

Heinemeier Hansson showed the local version at 5:47am EST Monday: "Omarchy on Nvidia N1X, running Nemotron 3.5 Bolt at 90 tps, diagnosing a sleep issue while offline, feels like the future." Tps is tokens per second. He blamed the N1X's rough reception on "a terrible out-of-the-box software experience. We can fix everything!" Omarchy 4.5 is "so, so close," he said, and he will be at Fireworks Forge in San Francisco on Tuesday to talk about "the agentic OS."

IV · WORK AND AGENTS

"Your job is to lead the agents," and Karpathy builds a site in two hours

Levie said every information job becomes managing a team of agents. Collison let users rewrite his apps by typing a request, and Heinemeier Hansson took a hard line on Proton's apology.

43M	6,454	150
Hacker News comments Claude filtered for Karpathy's HackerTube	Videos that survived, the ones Hacker News keeps recommending	Improvements one user made to a Collison app in a few days

Uncle Bob Martin @unclebobmartin · Author, Clean Code

Martin credited the line to Proton's chief technology officer at 8:47am EDT Saturday:

> *"Your job is not to use agents to help you write code. Your job is to lead the agents to help them write code."*
>
> UNCLE BOB MARTIN · @UNCLEBOBMARTIN · 8:47AM EDT · POST

Aaron Levie @levie · CEO, Box

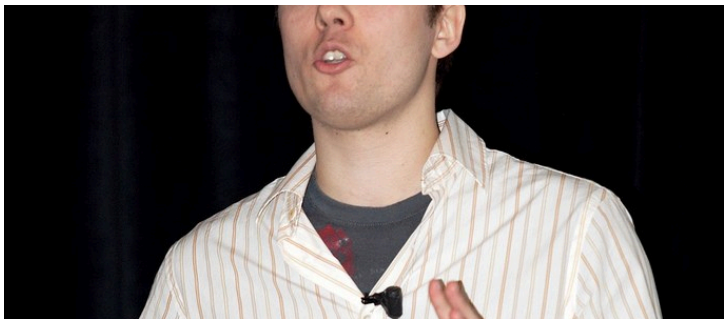
Levie quoted Martin at 1:01am EST Monday: "We're clearly moving from a world where agents help us do our work incrementally better or faster, to us helping agents do their work." That makes everyone in those roles "a manager of a team of agents," he wrote, in coding, cybersecurity, legal work, biotech and engineering, and output will be "decoupled from the normal ratio of hours spent working." On Saturday he offered a test: "if you could have done what you're doing before AI, you're likely not pushing it enough."

David Heinemeier Hansson @dhh · Creator, Ruby on Rails

Proton responded to criticism of its AI push Friday, saying it does not track how much AI code employees produce and will publish its AI policies and open-source internal tools. Heinemeier Hansson, at 4:05am EDT Saturday: "All very reasonable except promising no job losses due to obstinance on AI. Of course you should get fired if you refuse to comply with company strategy. That's the job!" Monday he wrote that "the great gift of agents is how they allow you to route around every insufferable gatekeeper in software." Omarchy's own bot merged 69 pull requests last week to his 73; he still reviews them through "the private Omabot Desk": "Omabot is preparing the brief, I make the decision."

ON THE WIRE

Bot versus founder. Omarchy's bot merged 69 pull requests last week, four fewer than David Heinemeier Hansson's 73.



David Heinemeier Hansson speaks at the O'Reilly Emerging Technology Conference in 2006.

PHOTO: JAMES DUNCAN DAVIDSON/O'REILLY MEDIA · CC BY 2.0 VIA WIKIMEDIA COMMONS

Patrick Collison @patrickc · Co-founder and CEO, Stripe

At 5:02pm EDT Saturday, Collison described an "improvements" box in his personal apps: "a freeform text input that's hooked up to an agent that yolo implements whatever is submitted." One user "has already implemented 150 improvements over the course of a few days." He called it "liquid software" and asked whether it stays maintainable, how it works for collaborative software, and whether it can be secure and reliable. It reminded him of Lisp machines, Smalltalk and HyperCard, "systems that the user can take control of and modify."

Andrej Karpathy @karpathy · Founder, Eureka Labs

> *"Weekend hack: HackerTube. I was annoyed with YouTube's recommendations so I had Claude filter down 43M comments over 20 years of HN to 6,454 videos that keep being recommended. *Significantly* better viewing experience, ~2 hours with Opus 5.5"*
>
> ANDREJ KARPATHY · @KARPATHY · 1:14PM EST · POST

HN is Hacker News. Claude first cut the pile with URLs, regular expressions and spam filters, and only the remainder went to the Sonnet model through a Claude Code workflow.

QUICK HIT

Suhail Doshi (@Suhail), founder of Particle, at 1:22pm EST Sunday: "I am looking for an AI agent engineer. Someone who will build open multi-agent systems." The job is in San Francisco only, working directly with him.

Tobi Lütke @tobi · CEO, Shopify

A Shopify engineer wrote that "moving to monorepo has been a boon for agents at shopify." Lütke recalled at 10:39am EST Sunday that a colleague warned him "every engineer absolutely hates" the idea, but once it's there, "no engineer would ever want to work at a company that doesn't use one." He also urged people to do holiday shopping "in bots this year", with the right skills installed. Told that scammers are targeting agent shoppers on the Shop app, he replied: "its a top priority for us."

V · ASSISTANTS AND PLATFORMS

Musk explains Grok Bot's usage limits as Sweeney presses Apple over CyberLink

Grok Bot's serving costs were too high, Musk said, as it gained gift buying and phone-screen help. Meta's Muse kept drawing praise, and Apple's suspension of a Taiwanese app maker became Sweeney's latest case.

$100	14.69%	0 of 11
X Money one user let Grok Bot spend on a surprise gift	Share of CyberLink's 2025 revenue from products pulled from Apple platforms	Devcon and Devconnect gatherings India had hosted before Mumbai

Elon Musk @elonmusk · CEO, Tesla and SpaceX

Asked Sunday for a Grok Bot usage reset because limits "ran out really fast this week," Musk replied at 3:34pm EST: "The cost of serving Grok Bot was unsustainably high relative to revenue." To a user locked out until Wednesday he asked: "Upgrade to higher usage plan cost increase is too high?" He promoted new abilities: Grok Bot "will research gifts that your friends and family would love and buy them," quoting a user who let the bot spend up to $100 from an X Money account, and it can now see what's on a phone screen. On the bot's pinned spot in the X inbox: "Just unpin it if you don't want it there."

> *"Try Grok @Bot! You will love it. We land a few dozen fixes, features and sometimes powers to Bot 7 days a week and we look at your feedback to make Bot as perfect as possible."*
>
> ELON MUSK · @ELONMUSK · 6:20PM EST · POST

He was quoting a Japanese user who called it "far better than Claude Code or Codex" (translated). Michael Truell (@mntruell), chief executive of Cursor, asked a user who had fully switched to Grok Bot: "How could it be improved?"

BY THE NUMBERS

220k	GB300 chips each in Macroharder and Minihard, per Elon Musk
6,454	Videos HackerTube kept from 43M Hacker News comments, per Andrej Karpathy
$1,536	New car insurance price Muse found against a $2,123 renewal, per Alexandr Wang
700+	Falcon 9 flights, per Elon Musk

Alexandr Wang @alexandr_wang · Chief AI officer, Meta

Wang quoted a user who said Meta's Muse moved his car insurance from a $2,123 renewal to $1,536 elsewhere: "if this isn't personal superintelligence i don't know what is." He also posted "grok might be the most truth-seeking ai after all" early Saturday, and on Sunday warned that anyone claiming to be Muse and asking for a phone number "is a phishing attack."

Tim Sweeney @TimSweeneyEpic · CEO, Epic Games

Apple stopped downloads of some CyberLink products on Oct. 24, saying they did not meet certain requirements. They made up 14.69% of the Taiwanese company's 2025 revenue, its investor presentation says, and CyberLink has appealed. Sweeney, at 3:28am EDT Saturday: "Apple is now blocking Cyberlink, Taiwan's #3 app company. That will certainly receive regulatory scrutiny." Earlier: "The walls are closing in on Apple's App Store monopoly." He is in Tokyo, where Unreal Fest starts this week.

QUICK HIT

Vitalik Buterin (@VitalikButerin), co-founder of Ethereum, arrived in Mumbai for Devcon on Saturday. India is 17% of the world, he wrote, and "has so far had 0 of 11 devcon(nect)s."

Sundar Pichai @sundarpichai · CEO, Google and Alphabet

Pichai posted "Hello Japan!" at 2:18pm EDT Friday, quoting Waymo's announcement that it will "scale a fleet of bZ4X autonomous vehicles in Japan", starting with testing and validation in 2027.

Paul Graham @paulg · Co-founder, Y Combinator

At 5:46pm EST Sunday: "Most investors have no idea of the damage that the mere distraction of talking to them does to a startup." He noted Monday that the deadline for Y Combinator's Winter 2027 batch is today.

COMING UP

Mon., Nov. 2 Application deadline for Y Combinator's Winter 2027 batch

Tue., Nov. 3 David Heinemeier Hansson talks about "the agentic OS" at Fireworks Forge in San Francisco

This week Unreal Fest in Tokyo

Mon., Nov. 9 Lame-duck session opens, where OpenAI plans its policy push, per Axios

Takeaways

1. Compute is scaling on the ground and testing in orbit: Colossus 2 targets about 1.2 million Blackwells, and Google ran its first prompt on a TPU in space.
2. GDPx shows long, high-value agent work is still mostly unsolved, with the best model at 12%.
3. Claude's rule against game remakes made tool control a live issue, and open, local models are the answer many builders give.
4. The new job description, Levie says, is managing a team of agents rather than working with an AI helper.
5. Grok Bot's usage limits reflect serving costs, Musk said, even as new abilities keep arriving.
6. Apple's suspension of CyberLink adds to pressure on its control of app distribution.

On the beat

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Elon Musk, Tim Sweeney, David Heinemeier Hansson, Guillermo Rauch, Palmer Luckey, Uncle Bob Martin, Paul Graham, Suhail Doshi, Yann LeCun, Albert Gu, Michael Dell, Tobi Lütke, Joe Gebbia, Alexandr Wang, Vitalik Buterin, François Chollet, Sundar Pichai, Andrej Karpathy, Jeff Dean, Patrick Collison, Fei-Fei Li, Kevin Weil, Demis Hassabis, Ashok Elluswamy, Kanjun Qiu, Aaron Levie, Aidan Gomez, Satya Nadella, John Carmack, Jim Fan, Mustafa Suleyman, Michael Truell, Lisa Su.

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Marc Andreessen, Jensen Huang, Sam Altman, Dario Amodei, Greg Brockman, Andrew Ng, Tom Brown, Clément Delangue, Mira Murati, Mark Zuckerberg, Noam Brown, David Singleton, Ilya Sutskever, Nat Friedman, Arthur Mensch, Guillaume Lample, Mark Chen.

The list stands at 50.

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